

ANALYSIS OF CATEGORICAL DATA

15.1 INTRODUCTION

The term *categorical data* refers to observations recorded either on a nominal scale or on a discrete ordinal scale (so ties are expected to occur often). Typical examples of the nominal scale occur when data represent frequencies of categories of some qualitative attribute (e.g., responses in a questionnaire about the state of residence or religious affiliation). The discrete ordinal scale consists of a set of naturally ordered categories. For example, opinion on a specific issue may be classified as “favorable,” “neutral,” or “unfavorable”; education achieved may be classified as “high school,” “junior college,” “four-year college or university,” “graduate school”; and so on.

A special case is played by a binary data, for example male vs. female, smoker vs. non-smoker. There is no “natural” ordering here, but one can always assign values 0 and 1 (or any other two values) to the categories.

The following examples illustrate some of the possible problems in the analysis of categorical data.

■ EXAMPLE 15.1

A typical case of data on a nominal scale occur in genetic experiments. Suppose that we have a gene with two forms, A and a , so that an individual (e.g., plant) belongs to one of the three categories AA , Aa , or aa . If none of the forms is dominant, the three genotypes can be identified (they coincide with the phenotypes). According to genetic theory, the probabilities of three genotypes are θ^2 , $2\theta(1 - \theta)$, and $(1 - \theta)^2$, respectively, where θ is the unknown frequency of allele A . Suppose that out of 200 plants we have 100 of type AA , 89 of type Aa , and 11 of type aa . Are these data in agreement with genetic theory?

■ EXAMPLE 15.2

Suppose that we have joint data on ethnicity and incidence of various types of cancer. In studying the association between these two attributes, we might search for genetic, dietary, or other reasons for a lack of independence.

In the last example, both variables (ethnicity and type of cancer) were of a nominal character. When both variables are measured on an ordinal scale, we may additionally be interested in the strength of association between variables. Such questions are discussed later in this chapter, where we present selected methods of analysis of ordinal aspects of categorical data.

■ EXAMPLE 15.3

For data measured originally on an ordinal scale and later grouped into classes, the methods of Chapter 14 are not applicable. Suppose that we want to study a relationship between level of education (classified as “I, no college”; “II, some college”; and “III, at least four-year college/university”) and the frequency of changing jobs (again classified as high, moderate, or low). The data take the form

Frequency of job change	Education		
	I	II	III
Low	N_{11}	N_{12}	N_{13}
Moderate	N_{21}	N_{22}	N_{23}
High	N_{31}	N_{32}	N_{33}

Here N_{ij} is the frequency of occurrence of a given category among randomly selected study participants. We want to test the hypothesis that there exists an association (positive or negative) between the frequency of changing jobs and the level of education.

The theory of categorical data is a domain with a long tradition, the chi-square test being one of the oldest examples of statistical procedures that use only frequencies. At the same time, this domain has been developing rapidly over the last four decades. We will introduce some of the basic information on problems and methods in analysis of categorical data. An exhaustive presentation of the field can be found in Agresti (2002).

15.2 CHI-SQUARE TESTS

In this section, we present one of the oldest and best known statistical tests. Beginning with the case of a discrete distribution with a finite number of values, we let X be a random variable with r possible values $x_1, \dots, x_r$ and the corresponding probabilities $p_i = P\{X = x_i\}$, $i = 1, \dots, r$. For a random sample $X_1, \dots, X_n$, counts $N_1, \dots, N_r$, where $N_1 + \dots + N_r = n$, are respective frequencies of values $x_1, \dots, x_r$. The vector $(N_1, \dots, N_{r-1})$ (notice that one of the coordinates N_i is redundant), called the *count vector* of the sample, carries all the necessary information about the distribution of X . Formally, the likelihood of the data is

$$L = C p_1^{N_1} p_2^{N_2} \cdots p_r^{N_r} = C p_1^{N_1} p_2^{N_2} \cdots p_{r-1}^{N_{r-1}} \left(1 - \sum_{i=1}^{r-1} p_i\right)^{n - \sum_{i=1}^{r-1} N_i},$$

where $C = n!/(N_1!N_2!\cdots N_r!)$. Consequently, the count vector is a set of jointly sufficient statistics for the vector $(p_1, p_2, \dots, p_{r-1})$.

An important remark here is that procedures for the inference about the distribution $(p_1, p_2, \dots, p_{r-1})$ of X on the basis of the counts $(N_1, \dots, N_{r-1})$ do not depend on $x_1, \dots, x_r$.

Suppose that we want to test the hypothesis

$$H_0 : p_i = p_i^0, \quad i = 1, \dots, r \quad (15.1)$$

against the general alternative

$$H_1 : \text{hypothesis } H_0 \text{ is not true.}$$

Here, $(p_1^0, \dots, p_r^0)$ is some fixed probability distribution, and we assume that $p_i^0 > 0$ for $i = 1, \dots, r$.

Under the null hypothesis, for each i , the marginal distribution of the count N_i is $\text{BIN}(n, p_i^0)$. The test will be based on the following theorem.

Theorem 15.2.1 *Let $(N_1, \dots, N_r)$ be the count vector of random sample of size n from multinomial distribution with probabilities $(p_1^0, \dots, p_r^0)$, so that $p_1^0 + \dots + p_r^0 = 1$ and $N_1 + \dots + N_r = n$. Then the statistic*

$$Q^2 = \sum_{j=1}^r \frac{(N_j - np_j^0)^2}{np_j^0} \quad (15.2)$$

has the limiting (as $n \rightarrow \infty$) chi-square distribution with $r - 1$ degrees of freedom. The statistic Q^2 is often referred to as Pearson's chi-square.

Proof: A simple argument shows that the theorem is true for $r = 2$. In this case we have, remembering that $p_2^0 = 1 - p_1^0$ and $N_2 = n - N_1$,

$$\begin{aligned} Q^2 &= \frac{(N_1 - np_1^0)^2}{np_1^0} + \frac{(N_2 - np_2^0)^2}{np_2^0} = \frac{(N_1 - np_1^0)^2}{np_1^0} + \frac{(n - N_1 - n(1 - p_1^0))^2}{n(1 - p_1^0)} \\ &= \frac{(N_1 - np_1^0)^2}{np_1^0} + \frac{(N_1 - np_1^0)^2}{n(1 - p_1^0)} = \left[\frac{N_1 - np_1^0}{\sqrt{np_1^0(1 - p_1^0)}} \right]^2 \approx Z^2 \sim \chi_1^2. \end{aligned}$$

For an arbitrary r , the algebra is more complicated: the statistic Q^2 is represented as a sum of $r - 1$ squares of random variables, each converging in distribution to a standard normal random variable, and such that their coefficients of correlation tend to 0 as n increases. We omit the details. $\square$

As a practical rule, one obtains a reasonable approximation of the distribution of Q^2 if expected counts np_i^0 , $i = 1, \dots, r$, are at least 5, and the approximation is good if the expected counts exceed 10.

To test the null hypothesis $H_0 : p_j = p_j^0$, $j = 1, \dots, r$ against the alternative $H_1 : \text{"hypothesis } H_0 \text{ is false,"}$ we need to determine the critical region. Since any violations of the null hypothesis in the chi-square test will tend to increase the value of statistic Q^2 , we should take the right tail as the critical region, with $\chi_{\alpha, r-1}^2$ as a critical value.

■ EXAMPLE 15.4

According to genetic theory, the seeds collected from a field of pink pea should produce plants with white, pink, and red flowers, in the proportion 1:2:1. Of 400 plants grown

from such seeds, 93 are white, 211 are pink, and 96 are red. Does this result contradict genetic theory?

SOLUTION. We have here 400 observations of a three-valued random variable. According to the null hypothesis, we have $p_1^0 = 1/4$, $p_2^0 = 1/2$, and $p_3^0 = 1/4$, so the expected counts np_i^0 are 100, 200, and 100. The observed value of the test statistic Q^2 is

$$Q^2 = \frac{(93 - 100)^2}{100} + \frac{(211 - 200)^2}{200} + \frac{(96 - 100)^2}{100} = 1.255.$$

Since we have here 2 degrees of freedom, the p -value is 0.534, and consequently, the data do not provide enough evidence against the null hypothesis.

As already mentioned, the chi-squared test can be used for testing the hypothesis that the data of a continuous type follow a specific distribution. In this case, to obtain the count vector $(N_1, \dots, N_r)$, we partition the range of values of the observed random variable X into r sets, say $C_1, C_2, \dots, C_r$. Typically, the sets C_j are intervals, but this is not necessary; the sets C_j need not to be connected, and may consist of a number of noncontiguous intervals. The count N_j , given the sample of n observations of random variable X , is defined as a frequency of observations in set C_j . Clearly, $N_1 + \dots + N_r = n$. If f is the density of X specified by the null hypothesis, then we have

$$p_j^0 = \int_{C_j} f(x) dx, \quad j = 1, \dots, r.$$

The test statistic Q^2 depends not only on the sample but also on the choice of the partition into sets C_j . Sometimes, the choice of the partition is natural, while in other cases, the partition is chosen in a rather arbitrary way.

■ EXAMPLE 15.5

Fox and James (1987) give the following data about the birth signs of 851 prominent chess players:

Capricorn	63	Lasker, Keres, Chiburdanidze
Aquarius	79	Spassky, Bronstein
Pisces	101	Fischer, Tarrasch, Geller, Larsen
Aries	76	Smyslov, Kasparov, Korchnoi, Portisch
Taurus	77	Miles, Nunn, Steinitz
Gemini	67	Petrosian, Karpov, Short, Euwe
Cancer	54	Morphy, Anderssen
Leo	67	Botvinnik
Virgo	63	Philidor
Libra	69	Rubinstein, Fine
Scorpio	71	Capablanca, Alekhine, Nimzowitsch, Tal
Sagittarius	64	Reshevsky, Pillsbury.

It seems that Pisces have a significantly higher number of prominent chess players.

We want to test the null hypothesis that the birthday is not related to chess talent. In this case, we can expect that the birthdays of $n = 851$ prominent chess players form a random sample from a distribution uniform on the year (which we may conveniently

regard as a continuous distribution). The expected count for each sign will be $851 \times (1/12) = 70.917$, and

$$Q^2 = \frac{(63 - 70.917)^2}{70.917} + \cdots + \frac{(64 - 70.917)^2}{70.917} = 21.53.$$

The p -value of the observed result is 0.028 which means that such a result will occur, on average, about once in 35 times, if the null hypothesis is true. Someone with a firm belief that there is some truth in astrology may use this result to support an argument. A sceptic who is convinced that the configuration of stars and planets at the time of one's birth cannot affect this person's talents will regard the observed result as an example of a type I error.

■ EXAMPLE 15.6

The left tail (indicating a good fit) of the chi-square test can be used for detecting whether the data were tampered with. The point is that the p -value gives the probability that in a repetition of the experiment, one would observe worse (in discrete case, no better) fit than the one actually observed. Thus, if the p -value is close to 1, say 0.99, it means that on average, only once in 100 repetitions one can expect to observe a better fit. That strongly suggests that the data were "improved" to make them conform better to the null hypothesis. An interesting example of such a type of inference is provided by Fisher's analysis of data on heredity by Gregor Mendel. For a detailed explanation, see Freedman et al. (1992). Here, we briefly sketch the idea and results.

Mendel studied the laws of inheritance of various characteristics, eventually introducing the concept of a gene. In a typical experiment of Mendel, plants of genotype AA are crossed with plants of genotype aa . All seeds are then hybrids Aa . A number of such seeds are grown and the plants are cross-pollinated. According to genetic theory, the ratio of plants of genotypes AA , Aa , and aa are 1:2:1. If AA and Aa cannot be distinguished, then the ratio of genotype aa to all others is 1:3, and so on. For example, in the last case, of 800 plants, about 200 can be expected to be aa . In all of Mendel's experiments, the observed numbers differ suspiciously little from the expected. For example, suppose that in the last case Mendel reported 205 plants with an aa genotype out of 800. The chi-square fit of such a result is

$$\frac{(205 - 200)^2}{200} + \frac{(595 - 600)^2}{600} = 0.167,$$

which corresponds to a p -value of about 90%.

A single result with so high a p -value is not unusual, but when Fisher combined all of Mendel's data using the combined chi-square test, the p -value was 0.99996 (i.e., only four times out of 100,000 one can expect a better fit). Thus, either Mendel had some extraordinary luck or his data were "beautified" to conform better to his theory.

In the cases considered above, the null hypothesis is simple; that is, the hypothetical distribution is specified completely. More often the null hypothesis comprises a class of distributions. We will now consider cases where the distribution of a random variable X depends on some parameter θ (possibly vector valued) in a specified family of distributions. Thus, we will assume that X is discrete random variable, with r possible values $x_1, \dots, x_r$, and such that

$$P\{X = x_j\} = p_j(\theta),$$

where $\theta = (\theta_1, \dots, \theta_k) \in H_k$. Moreover, we will assume that the number of classes r satisfies the inequality $r \geq k + 2$ and that $p_j(\theta) > 0$ for all j and all $\theta \in H_k$.

As before, we assume that we have a random sample of values of X , leading to the count vector $(N_1, \dots, N_r)$, where $N_1 + \dots + N_r = n$. This time we cannot form the test statistic (15.2) because the expected class frequencies $np_j(\theta)$ depend now on the unknown parameter θ . According to Theorem 15.2.2, we must use instead the estimated expected frequencies obtained as functions of the maximum likelihood estimate (MLE) of θ . Given the count $(N_1, N_2, \dots, N_r)$ the likelihood of the data is

$$L(\theta; N_1, \dots, N_r) = [p_1(\theta)]^{N_1} \dots [p_r(\theta)]^{N_r}. \quad (15.3)$$

Let $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_k)$ denote the value of the parameter θ that maximizes the likelihood (15.3). We have then the following theorem.

Theorem 15.2.2 *The statistic*

$$Q^2 = \sum_{j=1}^r \frac{[N_j - np_j(\hat{\theta})]^2}{np_j(\hat{\theta})} \quad (15.4)$$

has, as $n \rightarrow \infty$, the limiting chi-square distribution with $r - 1 - k$ degrees of freedom.

We omit the proof, which can be found in Cramér (1946).

■ EXAMPLE 15.7

Consider a gene with two alleles, A and a . Let the frequency of gene A in population be θ . Under random mating, the frequencies of individuals of genotypes AA , Aa , and aa are θ^2 , $2\theta(1 - \theta)$, and $(1 - \theta)^2$. To test the theory, n individuals are randomly selected, and the count of the three genotypes is N_1, N_2, N_3 . Then the likelihood of the data is

$$L = [\theta^2]^{N_1} [2\theta(1 - \theta)]^{N_2} [(1 - \theta)^2]^{N_3} = 2^{N_2} \theta^{2N_1 + N_2} (1 - \theta)^{N_2 + 2N_3}.$$

Differentiating $\log L$, we obtain easily the MLE of θ :

$$\hat{\theta} = \frac{2N_1 + N_2}{2n}. \quad (15.5)$$

For a numerical example, suppose that $n = 200$, $N_1 = 25$, $N_2 = 10$, and $N_3 = 165$. We have then $\hat{\theta} = 60/400 = 0.15$, and using (15.5), we get

$$Q^2 = \frac{(25 - 30)^2}{30} + \frac{(10 - 51)^2}{51} + \frac{(165 - 144.5)^2}{144.5} = 36.70.$$

Since this result exceeds $\chi_{0.0005,1}^2 = 7.879$, the p -value is less than 0.005, so the evidence against the null hypothesis of random mating provided by such data is very strong.

Theorem 15.2.1 asserts that if the null hypothesis is simple (specifies completely the distribution), then the limiting distribution of Q^2 (under null hypothesis) is chi-square with $r - 1$ degrees of freedom. On the other hand, if the null hypothesis is composite, and we have to estimate the expected counts by finding the MLE's of the parameters $\theta_1, \dots, \theta_k$ given the counts, we lose k degrees of freedom.

■ **EXAMPLE 15.8**

Assume that we have raw data for the numbers $x_{1905}, x_{1906}, \dots, x_{1991}$ of cloudless nights in the last $n = 87$ years at some prospective telescope site. Suppose that it is known that $x_{1905} + x_{1906} + \dots + x_{1991} = 21,163$ and $x_{1905}^2 + \dots + x_{1991}^2 = 5,226,819$. However, the actual data are not available, and instead we have the following counts N_i of years with given numbers of cloudless nights:

Interval	N_i
160 or below	1
161 to 180	3
181 to 200	7
201 to 220	17
221 to 240	18
241 to 260	26
261 to 280	9
281 to 300	4
301 or above	2

We want to test the hypothesis that the number of the cloudless nights at the site is normally distributed.

The MLE's of μ and σ^2 computed from original data are

$$\bar{x} = \frac{21,163}{87} = 243.25 \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{87}(5,226,819) - (243.25)^2 = 907.82;$$

hence $\hat{\sigma} = 30.13$. The estimated class probabilities are, letting X denote the number of cloudless nights,

$$p_1 = P(X \leq 160) = \Phi\left(\frac{160 - 243.25}{30.13}\right) = \Phi(-2.76) = 0.0029,$$

$$p_2 = P(160 \leq X \leq 180) = \Phi(-2.10) - \Phi(-2.76) = 0.015,$$

and similarly for subsequent intervals.

The actual and expected counts for the consecutive classes are therefore

Interval	$87p_i$	N_i
160 or below	0.252	1
161 to 180	1.305	3
181 to 200	4.959	7
201 to 220	12.676	17
221 to 240	19.192	18
241 to 260	22.281	26
261 to 280	15.356	9
281 to 300	7.056	4
301 or above	2.612	2

To avoid too small expected class sizes, we combine the first three classes together as well as the last two classes and obtain:

Interval	$87p_i$	N_i
200 or below	6.516	11
201 to 220	12.676	17
221 to 240	19.192	18
241 to 260	22.281	26
261 to 280	15.356	9
281 or above	9.668	6

The observed value of the statistic Q^2 is now 9.278. This corresponds to the p -value equal 0.026 (3 df).

PROBLEMS

- 15.2.1** Ladislaus von Bortkiewicz, a Russian economist and statistician, is known for the data he collected on the number of Prussian cavalryman being killed by the kick of a horse. He observed 10 army corps for 20 years, obtaining 200 observations. The total of 122 deaths was distributed as follows:

0	1	2	3	4	5	6
109	65	22	3	1	0	0

Test the hypothesis that the number of deaths from horse kicks has Poisson distribution.

- 15.2.2** A certain type of toy is sold with three batteries included. The number of defective batteries (X) in a random sample of 200 toys are as follows:

X	0	1	2	3
Count	51	92	40	17

Test the hypothesis that the number of defective batteries in a toy has a binomial distribution.

- 15.2.3** Assume that the sexes of children in a family are independent. In a human population, the probability that a child is a male is very close to 0.5. Numbers of boys (X) in a random sample of 100 families with four children are:

X	0	1	2	3	4
Count	7	21	40	27	5

Test the hypothesis that the distribution of the number of boys in a family of four children is indeed $\text{BIN}(4, 0.5)$.

- 15.2.4** Suppose that counts of female offsprings in a certain animal species with four offsprings are 3, 8, 28, 40, 21. Test the hypothesis that the corresponding distribution is binomial.
- 15.2.5** Given the data on numbers of hits of various areas of London by V2 rockets (see Example 8.14), test the hypothesis that the numbers of hits have a Poisson distribution.

15.3 HOMOGENEITY AND INDEPENDENCE

Chi-square tests can also be used for testing hypotheses about jointly distributed categorical variables. The data obtained in a random sample are summarized in a *contingency table* with r rows and c columns representing levels of respective classification variables. If n observations are taken, then N_{ij} is the number of observations classified at the i th level of the first classification variable and at the j th level of the second. The counts can be arranged into a matrix $[N_{ij}]$, where we have $\sum_{ij} N_{ij} = n$.

We will postpone a more detailed analysis of types of contingency tables to subsequent sections. Here we will consider only the simplest case, corresponding to random sampling from a population whose elements are categorized according to some classification system (gender vs. educational level, smoking status vs. cause of death, etc.).

Let p_{ij} be the probability that a single observation belongs to the i th category in first classification and j th category in the second classification. The marginal probabilities here are

$$p_{i+} = \sum_j p_{ij}, \quad p_{+j} = \sum_i p_{ij},$$

and the most obvious null hypothesis is that of independence of the two classifications:

$$H_0 : p_{ij} = p_{i+}p_{+j} \quad \text{for all } i, j.$$

The alternative hypothesis is

$$H_a : H_0 \text{ is not true.}$$

Let now $\hat{p}_{ij}$ denote the MLE of the probability p_{ij} based on the count matrix $[N_{ij}]$. Then, the test statistic (15.4) takes on the form

$$Q^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - n\hat{p}_{ij})^2}{n\hat{p}_{ij}}. \quad (15.6)$$

This statistic, according to Theorem 15.2.2, has a limiting (as $n \rightarrow \infty$) chi-square distribution with $rc - 1 - k$ degrees of freedom, where k is the number of estimated parameters.

To determine k , and also find the estimators $\hat{p}_{ij}$, observe that under the null hypothesis H_0 , we have (by the invariance property of MLE's) $\hat{p}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$. Now there are r values of the marginal probabilities p_{i+} , and c values of marginal probabilities p_{+j} , but in each of these marginal distributions, one value is a function of others, for instance,

$$p_{r+} = 1 - p_{1+} - p_{2+} - \cdots - p_{r-1,+}$$

and similarly

$$p_{+c} = 1 - p_{+1} - p_{+2} - \cdots - p_{+,c-1}.$$

Thus, the number of estimated parameters is $k = (r - 1) + (c - 1)$, and consequently the number of degrees of freedom of the limiting distribution of (15.6) is

$$rs - 1 - (r - 1) - (c - 1) = (r - 1)(c - 1). \quad (15.7)$$

Finally, we know that the MLE of a probability in multinomial distribution is the relative frequency, so

$$\hat{p}_{i+} = \frac{N_{i+}}{n}, \quad \hat{p}_{+j} = \frac{N_{+j}}{n},$$

where

$$N_{i+} = \sum_j N_{ij}, \quad N_{+j} = \sum_i N_{ij}. \quad (15.8)$$

Thus, $\hat{p}_{ij} = N_{ij}/n$, and substitution into (15.6) gives the following theorem.

Theorem 15.3.1 *Assume that n elements of a random sample are classified according to two attributes with r and c categories, respectively, producing a contingency table (count matrix) $[N_{ij}]$. For testing the hypothesis*

$$H_0 : \text{row and column classifications are independent}$$

against the general alternative

$$H_a : \text{hypothesis } H_0 \text{ is false,}$$

one can use the statistic

$$Q^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - N_{i+}N_{+j}/n)^2}{N_{i+}N_{+j}/n}, \quad (15.9)$$

which (under H_0) has limiting chi-square distribution with $(r - 1)(c - 1)$ degrees of freedom.

Any deviations from H_0 tend to increase Q^2 . So the critical region contains values of Q^2 that exceed $\chi^2_{\alpha, (r-1)(c-1)}$.

■ EXAMPLE 15.9

The Special Election Issue of *Newsweek* (November/December 1992) gives the exit poll results for the 1992 presidential election. One of the tables is the following:

		Clinton	Bush	Perot
White	87%	41%	38%	21%
Black	9%	82%	11%	7%

Although the conclusion seems quite clear, let us try to analyze this table and test the hypothesis that the preferences for the three candidates are independent of race. The total sample size given is $n = 15,241$ voters, and the margin of error is given as 1.1% points.

Observe first that the marginal percentages of whites and blacks do not add up to 100%, which means that the data for 4% of the voters (e.g., Hispanics) were not taken into account. Thus, we can estimate the sample size for our test to be about $0.96 \times 15,241 = 14,631$. Now the marginal totals for whites and blacks can be estimated as

$$N_{W+} = 14,631 \times \frac{0.87}{0.87 + 0.09} = 13,259,$$

and therefore $N_{B+} = 1,372$.

Next the percentages in each row add to 100%; this allows us to estimate the counts $N_{W,Clinton}$, $N_{W,Bush}$, and so on. The whole contingency table takes on the form

	Clinton	Bush	Perot	
White	5,436	5,039	2,784	13,259
Black	1,125	151	96	1,372
	6,561	5,190	2,880	14,631

We can now compute the observed value of statistic Q^2 given by formula (15.9). We obtain here $Q^2 = 845.2$, which exceeds the critical value for a chi-square distribution with 2 degrees of freedom chosen for any reasonable level of significance α . The race and voting preference are definitely dependent.

Continuing this analysis, we could ask what confidence level the pollsters use in announcing their “margin of error.” Here it is given as $\pm 1.1\%$ points. We are estimating here the true proportion p on the basis of sample of size $n = 14,631$. If X is the number of observations of a given category, then X has a binomial distribution with parameters n and p ; hence, X is approximately normal $N(np, np(1-p))$. Consequently, the estimated percentage, $100\hat{p} = 100X/n$, has an approximate normal distribution with mean $E(100\hat{p}) = 100p$ and $\text{Var}(100\hat{p}) = 100^2 \text{Var}(X/n) = 100^2 p(1-p)/n$. The $(1-\alpha)$ -confidence interval for the mean $100p$ is $100\hat{p} \pm z_{\alpha/2} \sqrt{\text{Var} 100\hat{p}}$.

The fact that the error given is the same for all data (instead of depending on the observed percentage) suggests that an upper bound $p(1-p) \leq 1/4$ is used. We have therefore the inequality

$$z_{\alpha/2} \sqrt{\frac{100^2 p(1-p)}{n}} \leq z_{\alpha/2} \frac{100}{\sqrt{4 \times 14,631}}.$$

The right-hand side equals 1.1 for $z_{\alpha/2} = 2.575$, which suggests that the pollsters used a 99% confidence level.

The chi-square test for independence described in Theorem 15.3.1 concerns the case where the counts N_{ij} arise from a cross-classification of independent and identically distributed observations. In experiments where one of the classification variables is not random but controlled by the experimenter, the data cannot be treated as a sample from a bivariate distributions.

■ EXAMPLE 15.10 Prospective and Retrospective Studies

Data summarized in contingency tables are usually used to analyze the hypothesized relationship between *cause* and *effect* (also referred to as *stimulus* and *response*, *explanatory* and *response* variable, *independent* and *dependent* variable, etc.). Let X and Y denote these variables, with X having r categories (also referred to as levels or *treatments*, depending on the context) and Y having c categories.

In *prospective studies*, one selects groups of subjects corresponding to various levels of X and then classifies each group separately according to levels of Y . For instance, in the social sciences, one may be interested in productivity (Y) and its dependence on stress level (X). The data may result from selecting r groups of subjects, exposing the i th group to the i th stress level, and then observing levels of productivity. Consequently, the totals N_{i+} (sizes of groups exposed to levels of stress) are not random. Similarly

in medical research, one may select two groups of patients and then administer the treatment being studied to one group and a placebo to the other group. The response Y can be observed in both groups. Again, the sizes of the groups are not random but are under the control of the experimenter.

In *retrospective studies* the situation is similar, except that now the marginal counts N_{+j} of the response categories are controlled by the experimenter. An example might be provided by typical data on smoking habits and lung cancer. A sample of subjects who died from lung cancer is selected and compared with a sample (possibly matched with respect to various attributes such as sex, age, etc.) of subjects who died from other causes. The sizes of these two samples are chosen largely at will. The two samples are then classified according to categories related to smoking.

For the analysis of contingency tables in which one of the marginal frequency vectors is fixed, let us first introduce the appropriate notation and then formulate the hypothesis to be tested.

The data form, as before, a count matrix $[N_{ij}]$, with marginal counts $N_{i+} = \sum_j N_{ij}$ and $N_{+j} = \sum_i N_{ij}$. The total number of observations is $n = \sum_{ij} N_{ij}$. For the sake of argument, assume that the counts N_{i+} , $i = 1, \dots, r$, are not random. For each i , the vector $(N_{i1}, N_{i2}, \dots, N_{ic})$ is assumed to represent the counts from N_{i+} iid observations, sampled from the distribution corresponding to the i th level of the first attribute. The probabilities in this distribution will be denoted by $(p_{1|i}, p_{2|i}, \dots, p_{c|i})$, where $p_{1|i} + p_{2|i} + \dots + p_{c|i} = 1$. Here, $p_{j|i}$ stands for the probability that the observation will fall into the j th class of the second attribute, if the sample is taken from the population of objects with the i th level of the first attribute. Note that $p_{j|i}$ is not a conditional probability as long as i is not random. We use, however, the symbols appropriate for conditional probabilities, since in the special case where i is random, we have the obvious relation

$$p_{ij} = p_{j|i}p_{i+}. \quad (15.10)$$

The hypothesis of interest here is the hypothesis of homogeneity, which may be stated as follows:

$$H_0 : \text{distributions } (p_{1|i}, \dots, p_{c|i}) \text{ do not depend on } i; \quad (15.11)$$

that is, for each j , we have $p_{j|1} = \dots = p_{j|r} = p_j$.

As before, we will obtain a test for H_0 against the general alternative

$$H_1 : \text{hypothesis } H_0 \text{ is not true.}$$

Despite the differences between the independence hypothesis and the homogeneity hypothesis above, they are tested by the same statistic. Indeed, for any fixed i , the component of the chi-square sum is

$$X_i^2 = \sum_{j=1}^c \frac{(N_{ij} - N_{i+}\hat{p}_{j|i})^2}{N_{i+}\hat{p}_{j|i}},$$

where $\hat{p}_{j|i}$ is the MLE of probability $p_{j|i}$. Clearly, under the null hypothesis of homogeneity we have

$$X_i^2 = \sum_{j=1}^c \frac{(N_{ij} - N_{i+}\hat{p}_j)^2}{N_{i+}\hat{p}_j}, \quad (15.12)$$

and $\hat{p}_j = N_{+j}/n$ (i.e., the MLE of p_j is the relative frequency of the j th category). Adding over i , we obtain the test statistic

$$Q^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - N_{i+}N_{+j}/n)^2}{N_{i+}N_{+j}/n}. \quad (15.13)$$

The number of degrees of freedom equals $r(c-1) - k$, where k is the number of estimated parameters. Indeed, each Q_i^2 will have $c-1$ degrees of freedom if the value p_j is known. The parameters estimated are $p_1, \dots, p_{c-1}$, so $k = c-1$, and we obtain the following theorem:

Theorem 15.3.2 *If the null hypothesis H_0 (15.11) is true, and if $N_{i+} \rightarrow \infty$ for $i = 1, \dots, r$, then the statistic Q^2 given by (15.13) has the limiting chi-square distribution with $r(c-1) - (c-1) = (r-1)(c-1)$ degrees of freedom.*

Any violation of the null hypothesis will tend to increase the value of Q^2 . So, again, the null hypothesis will be rejected if the observed value of the statistic Q^2 exceeds $\chi_{\alpha, (r-1)(c-1)}^2$.

PROBLEMS

15.3.1 Show that in the case of a 2×2 contingency table, the statistic Q^2 given by (15.9) is proportional to $(N_{11}N_{22} - N_{21}N_{12})^2$, and find the proportionality constant.

15.3.2 For each 3×4 contingency table below find such k that the hypothesis about independence of two classification variables is rejected at the 0.05 significance level.

(i)	k	k	k	k	(ii)	5	5	5	5
	k	k	0	k		5	5	k	5
	k	k	k	k		5	5	5	5

15.3.3 Mrs. Smith, who teaches an elementary statistics course, classified each student in the class according to whether the grade on the first exam was below or above the median for this exam, and then did the same for the second exam. The results obtained are as follows:

First exam	Second exam	
	Below	Above
Below	30	5
Above	5	30

Compute Q^2 and find the p -value. What legitimate conclusion can be made?

15.3.4 Professionals from various disciplines participated in a study on job-related stress. A random sample of size 100 was selected from each group of professionals (physicians, engineers, and lawyers), and each person was asked to evaluate the level of job-related stress as low, moderate, or high. The results of a study are given below.

	L	M	H
Physicians	5	25	70
Engineers	25	25	50
Lawyers	10	30	60

Specify the hypothesis to be tested, perform the test, and make the appropriate conclusions.

15.3.5 A random sample of 29 university students was selected and each student was then classified according to their high school GPA and college GPA. Both classifications

had the same two categories: “I, below 3.0,” “II, at least 3.0.” Formulate a hypothesis to be tested. Perform the test and make appropriate conclusions.

High school GPA	College GPA	
	I	II
I	5	3
II	12	9

15.4 CONSISTENCY AND POWER

Chi-square tests of either independence or homogeneity serve as tests against a general alternative, asserting simply that the null hypothesis is false (such tests are called *omnibus tests*). If the independence hypothesis is not true, then (as the sample size increases) the probability of rejection of the null hypothesis tends to 1. We may rephrase this property by stating that power of the chi-square test for independence tends to 1 for any simple hypothesis contained in the alternative as the sample size increases. This property of the test is called *consistency*.

The situation is similar in the case of tests for homogeneity, except that now the sample sizes refer to rows (levels of explanatory variable), and are determined not by chance but by the experimenter. Again, if the null hypothesis is not true, then at least two of the rows of the matrix $[p_{ij}]$ are different. The power of the chi-square test will tend to 1 on a simple alternative in which the rows labeled i_0 and i'_0 are different, if the sample sizes N_{i_0+} and $N_{i'_0+}$ both tend to infinity.

In general, it is difficult to analyze the power of chi-square tests for independence or homogeneity, since it involves determining the exact or limiting distribution of a statistic under some distribution in the underlying population.

However, if one looks at the main motivation of an analysis of the power of a test, namely to decide “which hypothesis should be accepted if one rejects null hypothesis,” then one can suggest the following approach.

When the null hypothesis is not valid, it typically is due to the fact that there is a strong association between some specific values of the two attributes analyzed, while for other values the null hypothesis is, at least approximately, satisfied.

In symbols, the null hypothesis (of independence) asserts that all absolute differences

$$|p_{ij} - p_{i+}p_{+j}| \quad (15.14)$$

are zero. If the null hypothesis is not valid, then there exist absolute differences that are positive. What typically happens in such cases is that a few of those differences are high; other may be close to zero (rather than all of these differences being small). In these cases, one would like to identify the cells for which the differences (15.14) are high.

To present the solution, let us first derive an alternative to the chi-square test, namely the generalized likelihood ratio (GLR) test (see Section 12.6). Consider the case of testing the hypothesis of independence, which states that $p_{ij} = p_{i+}p_{+j}$ for all i, j . The union of H_0 and H_1 allows the probabilities p_{ij} to be arbitrary (subject only to the constraint that $\sum_{i,j} p_{ij} = 1$). The data form the contingency table $[N_{ij}]$.

The likelihood of the data is (up to a multiplicative constant) equal to $L = \prod_i \prod_j (p_{ij})^{N_{ij}}$. It is maximized at $\hat{p}_{ij} = N_{ij}/n$, where $n = \sum_{i,j} N_{ij}$ is the total sample size. Consequently, the denominator in the GLR, equal to the maximum over all parameter space, is

$$\sup_{H_0 \cup H_1} L = \prod_i \prod_j \frac{(N_{ij})^{N_{ij}}}{n^{N_{ij}}} = n^{-n} \prod_i \prod_j (N_{ij})^{N_{ij}}.$$

On the other hand, the likelihood over the null hypothesis equals (up to the same multiplicative constant)

$$L = \prod_i \prod_j (p_{i+})^{N_{ij}} (p_{+j})^{N_{ij}}.$$

This is maximized at $\hat{p}_{i+} = N_{i+}/n$ and $\hat{p}_{+j} = N_{+j}/n$, and we obtain

$$\begin{aligned} \max_{H_0} L &= \prod_i \left(\frac{N_{i+}}{n} \right)^{N_{i+}} \prod_j \left(\frac{N_{+j}}{n} \right)^{N_{+j}} \\ &= n^{-2n} \prod_i \prod_j (N_{i+})^{N_{ij}} (N_{+j})^{N_{ij}}. \end{aligned}$$

The GLR equals therefore

$$\lambda = \frac{\prod_i (N_{i+})^{N_{i+}} \prod_j (N_{+j})^{N_{+j}}}{n^n \prod_i \prod_j (N_{ij})^{N_{ij}}} = \prod_i \prod_j \left(\frac{N_{i+} N_{+j}}{n N_{ij}} \right)^{N_{ij}}. \quad (15.15)$$

Under the null hypothesis H_0 , the statistic

$$G^2 = -2 \log \lambda = 2 \sum_{i=1}^r \sum_{j=1}^c N_{ij} \log \frac{N_{ij}}{(N_{i+} N_{+j})/n} \quad (15.16)$$

has a limiting (as $n \rightarrow \infty$) chi-square distribution with a number of degrees of freedom equal to the difference in the number of estimated parameters in the denominator and in the numerator. Thus, the number of degrees of freedom is

$$(rc - 1) - [(r - 1) + (c - 1)] = (r - 1)(c - 1).$$

Symbolically, we may write $Q^2 = \sum (N_{ij} - E(N_{ij}))^2 / E(N_{ij})$, where N_{ij} and $E(N_{ij})$ stand for observed and expected (under null hypothesis) counts. On the other hand, using the fact that

$$\log(1 + x) = x + \frac{x^2}{2} + \cdots,$$

we can write

$$\begin{aligned} G^2 &= 2 \sum_{i,j} N_{ij} \log \left(\frac{N_{ij}}{E(N_{ij})} \right) = 2 \sum_{i,j} N_{ij} \log \left(1 + \frac{N_{ij} - E(N_{ij})}{E(N_{ij})} \right) \\ &\approx 2 \sum_{i,j} N_{ij} \frac{N_{ij} - E(N_{ij})}{E(N_{ij})} + \sum_{i,j} \frac{(N_{ij} - E(N_{ij}))^2}{E(N_{ij})} \times \frac{N_{ij}}{E(N_{ij})}. \end{aligned}$$

Now, as $n \rightarrow \infty$, we have $N_{ij}/E(N_{ij}) \rightarrow 1$ in probability, and also $E[\sum (N_{ij} - E(N_{ij}))] = 0$. This suggests that the first sum is close to 0, and the second sum is close to Q^2 .

Although the argument above falls short of being a proof, it suggests that Q^2 and G^2 are close to one another for large samples. In fact one can show that $Q^2 - G^2 \rightarrow 0$ in probability as $n \rightarrow \infty$. The idea of using the statistic G^2 to investigate the power lies in the additivity property of the chi-square distribution. If W has χ_k^2 distribution, and $k = k_1 + k_2 + \cdots + k_s$ is a sum of positive integers, then there exist independent random variables $Y_1, \dots, Y_s$ such that $W = Y_1 + \cdots + Y_s$ and $Y_i \sim \chi_{k_i}^2$.

In the case under consideration, we have the random variable G^2 given by (15.16), which (under the null hypothesis) has a limiting chi-square distribution with $(r - 1)(c - 1)$ degrees

of freedom. It may be shown that G^2 can be represented as a sum of independent random variables $G_1^2, \dots, G_s^2$, each corresponding to a subtable of the original contingency table.

The subtables are obtained by taking a part of the original table and then collapsing some of the categories. Goodman (1969, 1971) and Lancaster (1949, 1969) formulated following necessary conditions for components G_i^2 in the sum

$$G^2 = G_1^2 + G_2^2 + \dots + G_s^2 \quad (15.17)$$

are independent:

1. The degrees of freedom for the components $G_1^2, \dots, G_s^2$ must sum to the number $(r-1)(c-1)$ of the degrees of freedom of G^2 .
2. Each cell count N_{ij} of the original table must appear in exactly one subtable.
3. Each of the marginal counts N_{i+} and N_{+j} of the original table must appear as marginal count in exactly one subtable.

If all three conditions are satisfied, then the values of statistic G^2 computed for the whole table and for the subtables satisfy (15.17).

■ EXAMPLE 15.11

Consider a 3×3 contingency table

	Y_1	Y_2	Y_3	Marginals
X_1	N_{11}	N_{12}	N_{13}	N_{1+}
X_2	N_{21}	N_{22}	N_{23}	N_{2+}
X_3	N_{31}	N_{32}	N_{33}	N_{3+}
Marginals	N_{+1}	N_{+2}	N_{+3}	

An example of a decomposition satisfying the three conditions of independence is as follows:

Subtable 1:

	Y_1	Y_2	
X_1	$\mathbf{N}_{11}$	$\mathbf{N}_{12}$	$N_{11} + N_{12}$
X_2	$\mathbf{N}_{21}$	$\mathbf{N}_{22}$	$N_{21} + N_{22}$
	$N_{11} + N_{21}$	$N_{12} + N_{22}$	

Subtable 2:

	$Y_1 \text{ or } Y_2$	Y_3	
X_1	$N_{11} + N_{12}$	$\mathbf{N}_{13}$	$\mathbf{N}_{1+}$
X_2	$N_{21} + N_{22}$	$\mathbf{N}_{23}$	$\mathbf{N}_{2+}$
	$N_{11} + N_{12}$ $+ N_{21} + N_{22}$	$N_{13} + N_{23}$	

Subtable 3:

	Y_1	Y_2	
X_1 or X_2	$N_{11} + N_{21}$	$N_{12} + N_{22}$	$N_{11} + N_{12}$ $+ N_{21} + N_{22}$
X_3	N_{31}	N_{32}	$N_{31} + N_{32}$
	N_{+1}	N_{+2}	

Subtable 4:

	Y_1 or Y_2	Y_3	
X_1 or X_2	$N_{11} + N_{12}$ $+ N_{21} + N_{22}$	$N_{13} + N_{23}$	$N_{1+} + N_{2+}$
X_3	$N_{31} + N_{32}$	N_{33}	N_{3+}
	$N_{+1} + N_{+2}$	N_{+3}	

The number of degrees of freedom of the original table is 4, and for each of the subtables it is 1, so the first condition is met.

Regarding conditions 2 and 3, we can check that they are met by simple inspection. In the four subtables, all entries and marginals from the original table are identified by boldface.

The algebraic verification that the condition (15.17) holds can be found in Lancaster (1949). Here we observe only that the partition of this example is not the only one possible for a 3×3 table: There are eight other partitions.

■ **EXAMPLE 15.12**

In a study of marijuana use in colleges, 445 students were sampled and classified according to a response variable (use of marijuana or other drugs) into three categories: “never,” “occasionally,” “regularly.” As a possible explanatory variable, the experimenters selected the number of parents who were alcohol or drug users (“neither one,” “exactly one,” “both”). The data that follow are from Devore (1991):

Parents’ drug use	Student’s drug use			
	Never (N)	Occasionally (O)	Regularly (R)	
None	141	54	40	235
One	68	44	51	163
Both	17	11	19	47
	226	109	110	445

For 3×3 tables, the statistic G^2 has a limiting chi-square distribution with 4 degrees of freedom. The value for the table above is $G^2 = 22.254$, which is highly significant (p -value equals 0.00018). Incidentally, for this table we have $Q^2 = 22.394$, which illustrates the closeness of G^2 and Q^2 for large samples. Thus, we may conclude that there is a relationship between parental and student use of drugs.

One of the questions we could ask here is whether a positive or a negative example is stronger. In other words, taking for granted that the frequency of marijuana use

tends to increase with the number of parents who use alcohol or drugs, the question is whether the effect of a bad example of one parent tends to outweigh the good example of the other parent. To get an insight into this question, we consider two decompositions of G^2 . The first decomposition correspond to the following four subtables:

	1		2			3		4	
	<i>N</i>	<i>O</i>	<i>N + O</i>	<i>R</i>		<i>N</i>	<i>O</i>	<i>N + O</i>	<i>R</i>
None	141	54	195	40	At most one	209	98	307	91
One	68	44	112	51	Both	17	11	28	19

The values of G^2 for these four tables are, respectively, $G_1^2 = 4.344$, $G_2^2 = 10.957$, $G_3^2 = 0.616$, and $G_4^2 = 6.336$, and we check that $G^2 = G_1^2 + G_2^2 + G_3^2 + G_4^2$. The corresponding p -values for 1 degree of freedom are, respectively, 0.037, 0.001, 0.432, and 0.012.

An alternative decomposition is as follows:

	1*		2*			3*		4*	
	<i>O</i>	<i>R</i>	<i>N</i>	<i>O + R</i>		<i>N</i>	<i>O</i>	<i>N + O</i>	<i>R</i>
One	44	51	68	95	Neither	54	40	141	94
Both	11	19	17	30	At least one	55	70	85	125

Now the values are $G_{1*}^2 = 0.871$, $G_{2*}^2 = 0.470$, $G_{3*}^2 = 3.893$, and $G_{4*}^2 = 17.019$. The corresponding p -values are 0.351, 0.493, 0.048, and 0.000.

These results indicate that a bad example, of just one parent, has a big influence on a student's marijuana use. As subtables 1* and 2* show, if at least one parent uses drugs or alcohol, then it does not really matter whether the other parent does as well. On the other hand, as subtables 3* and 4* show, there is a significant difference where none of the parents use drugs or alcohol. This effect increases greatly the likelihood that a student will never use the marijuana, and—if he uses it—at a decreasing frequency of use.

The conclusions are strengthened if one analyzes also a first decomposition. Here the category “one” is either compared with “neither,” or category “at most one” is compared with category “both.” As can be seen, subtables 1, 2, and 4 show the significant effect of a parent using alcohol or drugs. Subtable 3 shows that any positive effect of the other parent is limited. The presence or absence of a positive model does not have any significant effect on the frequency of marijuana use as long as this frequency is low (or zero). Subtable 4 suggests that the presence of a positive parent role model has only the effect of lowering the probability of a regular use of marijuana during the college years.

PROBLEMS

15.4.1 Find a decomposition of G^2 into independent components by decomposing: (i) A $2 \times k$ table. (ii) A 3×4 table.

15.4.2 The data on the incidence of a certain disease, classified by age and gender, are as follows:

	<20	20–39	40–59	60+
Men	10	15	30	40
Women	20	30	60	300

Find the values of statistics Q^2 and G^2 . Find the decomposition of G^2 , and verify that the “source” for lack of independence is the very high incidence of this disease among women over 60.

- 15.4.3** At the beginning of the semester a random sample of 104 students was selected out of students in all introductory statistics classes. Students were then classified according to their GPA (I, “below 3.0”; II, “between 3.0 and 3.5”; and III, “above 3.5”) and their attitude toward the statistics course (i, “I hate to take this class but I have to”; ii, “I do not mind taking this class but it is not one of my favorites”; iii, “I look forward to taking this class”). The results of classifications are given below.

	i	ii	iii
I	14	8	2
II	10	14	12
III	6	16	22

Test the hypothesis of independence of both classification variables. In the case of lack of independence, use decomposition to identify the cause of dependence.

15.5 2 × 2 CONTINGENCY TABLES

We begin with the simplest, but very common situation, where both variables have only two categories (i.e., are treated as binary variables). Denoting those levels by 1 and 2; we have then two distributions of the response, corresponding to the values of the explanatory variable, namely

$$(p_{1|1}, p_{2|1}) \quad \text{and} \quad (p_{1|2}, p_{2|2}),$$

where $p_{2|1} = 1 - p_{1|1}$ and $p_{2|2} = 1 - p_{1|2}$.

The null hypothesis of homogeneity reduces to

$$H_0 : \frac{p_{1|1}}{p_{1|2}} = 1.$$

When response 1 is in some sense undesirable (death, relapse of disease, etc.), the ratio $p_{1|1}/p_{1|2}$ was often called the *relative risk*. This term is now used generally regardless of the context (similarly as “success” and “failure” in a binomial distribution).

The alternative hypothesis of a positive association asserts that a higher value of explanatory variable gives a higher probability of a higher value of the response variable, that is, $p_{2|1} < p_{2|2}$. The latter inequality is equivalent to $1 - p_{1|1} < 1 - p_{1|2}$; hence $p_{1|1}/p_{1|2} > 1$. Similarly, a negative association means that the relative risk is less than 1.

An alternative formulation, which also suggests a testing procedure, is as follows: Consider a binary distribution $(\pi, 1 - \pi)$, where π is the probability of some event A . Then the ratio

$$\eta = \frac{\pi}{1 - \pi} \tag{15.18}$$

is called the *odds* (for the event A). It is clear that η determines π uniquely, namely $\pi = \eta/(1 + \eta)$. As π increases from 0 to 1, the odds η increase from zero to infinity. The odds

for the complement of the event A are $1/\eta$. In the case of the distributions $(p_{1|1}, p_{2|1})$ and $(p_{1|2}, p_{2|2})$, the odds (for the response 1) are

$$\eta_1 = \frac{p_{1|1}}{p_{2|1}} \quad \text{and} \quad \eta_2 = \frac{p_{1|2}}{p_{2|2}}.$$

To formulate the null and alternative hypotheses, it appears natural to consider the *odds ratio*:

$$\theta = \frac{\eta_1}{\eta_2} = \frac{p_{1|1}p_{2|2}}{p_{1|2}p_{2|1}} = \frac{p_{11}p_{22}}{p_{12}p_{21}}, \quad (15.19)$$

where the latter expression is meaningful in the case where the explanatory variable is random.

The null hypothesis, of either homogeneity or independence, has the form

$$H_0 : \theta = 1. \quad (15.20)$$

The alternative H_1^+ of positive and H_1^- of negative association are, respectively,

$$H_1^+ : \theta > 1 \quad \text{and} \quad H_1^- : \theta < 1. \quad (15.21)$$

Suppose now that the null hypothesis $H_0 : \theta = 1$ is to be tested against the alternative of positive association $H_a^+ : \theta > 1$. The data have the form of a 2×2 count matrix (contingency table):

$$\begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix}.$$

The question is: How to determine the p -value of the observed contingency table, that is, the *probability* (calculated under the assumption that the null hypothesis is true) *of observing—if the experiment were to be repeated—a contingency table that will at least be as much in favor of the alternative as the contingency table actually observed*.

To implement the idea of determining the p -value of the observed result, one needs to proceed as follows:

1. Explicate the ordering of the contingency tables according to the relation of “being more in favor of the alternative hypothesis.”
2. Specify the probabilities of occurrence of the various contingency tables under the null hypothesis. (Note that the null hypothesis of independence is composite; hence it does not lead directly to numerical values of probabilities.)

Regarding step 1, as a statistic, one can take a sample counterpart of the odds ratio θ , replacing the probabilities p_{ij} by the corresponding relative frequencies N_{ij}/n , where n is the total sample size. This leads to the statistic U , defined as

$$U = \frac{N_{11}N_{22}}{N_{12}N_{21}}$$

(and $U = \infty$ if N_{12} or N_{21} is zero).

Thus, if the observed contingency table is $[n_{ij}]$, then the p -value of this result is defined formally as

$$P_0 \left\{ [N_{ij}] : \frac{N_{11}N_{22}}{N_{12}N_{21}} \geq \frac{n_{11}n_{22}}{n_{12}n_{21}} \right\}. \quad (15.22)$$

It remains to specify the probability distribution P_0 on a suitably selected class of 2×2 contingency tables. The main requirement is that P_0 should not depend on any parameters

so that its numerical value can be determined for every contingency table for which P_0 is defined.

Now, if we take a random sample of size n from a population whose elements are classified according to two dichotomous classifications, then the probability of a particular contingency table with its sum of entries equal n is given by the multinomial distribution

$$P\{N_{ij} = n_{ij}, i, j = 1, 2\} = \frac{n!}{n_{11}!n_{12}!n_{21}!n_{22}!} p_{11}^{n_{11}} p_{12}^{n_{12}} p_{21}^{n_{21}} p_{22}^{n_{22}}. \quad (15.23)$$

If the null hypothesis H_0 is true, then $p_{ij} = p_{i+}p_{+j}$, and substitution into (15.23) gives

$$P_{H_0}\{N_{ij} = n_{ij}, i, j = 1, 2\} = \frac{n!}{n_{11}!n_{12}!n_{21}!n_{22}!} p_{1+}^{n_{1+}} p_{2+}^{n_{2+}} p_{+1}^{n_{+1}} p_{+2}^{n_{+2}}, \quad (15.24)$$

where $n_{i+} = n_{i1} + n_{i2}$ and $n_{+j} = n_{1j} + n_{2j}$ for $i, j = 1, 2$.

It should be clear that in order to obtain the probabilities of the contingency tables that do not depend on the parameter, it is necessary to restrict the definition of P_0 to the class of tables with given marginals n_{i+} and n_{+j} ($i, j = 1, 2$). This suggests taking as P_0 the conditional probabilities given both marginals, since the product $p_{1+}^{n_{1+}} p_{2+}^{n_{2+}} p_{+1}^{n_{+1}} p_{+2}^{n_{+2}}$ will then cancel.

We have, for the probability of marginals being (n_{1+}, n_{2+}) and (n_{+1}, n_{+2}) , the product of two binomial probabilities:

$$P\{N_{1+} = n_{1+}\} = \binom{n}{n_{1+}} p_{1+}^{n_{1+}} (1 - p_{1+})^{n - n_{1+}} = \binom{n}{n_{1+}} p_{1+}^{n_{1+}} p_{2+}^{n_{2+}}$$

and

$$P\{N_{+1} = n_{+1}\} = \binom{n}{n_{+1}} p_{+1}^{n_{+1}} (1 - p_{+1})^{n - n_{+1}} = \binom{n}{n_{+1}} p_{+1}^{n_{+1}} p_{+2}^{n_{+2}}.$$

Consequently,

$$\begin{aligned} P\{N_{1+} = n_{1+}, N_{+1} = n_{+1}\} &= P\{N_{1+} = n_{1+}\} P\{N_{+1} = n_{+1}\} \\ &= \binom{n}{n_{1+}} \binom{n}{n_{+1}} p_{1+}^{n_{1+}} p_{2+}^{n_{2+}} p_{+1}^{n_{+1}} p_{+2}^{n_{+2}}. \end{aligned} \quad (15.25)$$

Dividing (15.24) by (15.25), we obtain, after rearranging the multinomial coefficients, the following theorem in which $N_{ij}(x)$ is the 2×2 table with given marginals and $N_{11} = x$:

Theorem 15.5.1 *Under the null hypothesis $H_0 : \theta = 1$, for any integers n, n_{1+}, n_{+1} satisfying the conditions $n > 0, 0 \leq n_{1+} \leq n, 0 \leq n_{+1} \leq n$,*

$$\begin{aligned} P\{[N_{ij}] = [N_{ij}(x)] | N_{1+} = n_{1+}, N_{+1} = n_{+1}\} &= \frac{\binom{n+1}{x} \binom{n-n_{1+}}{n_{1+}-x}}{\binom{n}{n_{1+}}} \\ &= \frac{\binom{n+1}{x} \binom{n-n_{1+}}{n_{+1}-x}}{\binom{n}{n_{+1}}}. \end{aligned} \quad (15.26)$$

Observe now that we have

$$\frac{n_{11}n_{22}}{n_{12}n_{21}} = \frac{x(n - n_{1+} - n_{+1} + x)}{(n_{1+} - x)(n_{+1} - x)},$$

which is an increasing function of x . This means that the ordering of contingency tables with the same marginal totals coincides with an ordering with respect to the element $x = n_{11}$. Consequently, in view of Theorem 15.5.1, the p -value defined by (15.22) becomes

$$\sum_{x \geq n_{11}} \frac{\binom{n_{1+}}{x} \binom{n - n_{1+}}{n_{+1} - x}}{\binom{n}{n_{+1}}}.$$

■ EXAMPLE 15.13

To study whether or not there exists a positive association between musical and mathematical abilities, a group of 12 fourth graders was classified according to their scores (high or low) in these subject areas. The results are as follows:

Music	Mathematics		
	High	Low	
High	4	1	5
Low	2	5	7
	6	6	12

Do the observed data lead to rejecting the null hypothesis of independence of musical and mathematical abilities in favor of the alternative of a positive association?

SOLUTION. There is only one contingency table with the same marginals as original one, which is more in favor of the alternative, namely

5	0	5
1	6	7
6	6	12

The p -value of the observed result is therefore the sum of probabilities of both tables; hence it is

$$\frac{\binom{5}{4} \binom{7}{2}}{\binom{12}{6}} + \frac{\binom{5}{5} \binom{7}{1}}{\binom{12}{6}} = 0.121.$$

Clearly, at level 0.1 (and therefore also at any lower level), the null hypothesis should not be rejected: There is about a 12% chance of observing a result at least as much in favor of the alternative hypothesis if only due to random fluctuations (i.e., if in fact the null hypothesis is true).

Let us mention here that the test as described above is applicable also to the cases of prospective or retrospective studies where one of the marginals is not random. Indeed, if the marginals n_{1+} and n_{2+} are selected by the experimenter, then the contingency table is

n_{11}	n_{12}	a
n_{21}	n_{22}	b
n_{+1}	n_{+2}	n

(where we let a and b denote the nonrandom marginals to distinguish them from the random marginals). The corresponding table of conditional probabilities, under the null hypothesis of homogeneity, is

$$\begin{bmatrix} p_{1|1} & p_{2|1} \\ p_{1|2} & p_{2|2} \end{bmatrix} = \begin{bmatrix} \gamma & 1 - \gamma \\ \gamma & 1 - \gamma \end{bmatrix}.$$

The likelihood of the data, under the null hypothesis, is therefore a product of two binomial probabilities:

$$\begin{aligned} & \binom{a}{n_{11}} \gamma^{n_{11}} (1 - \gamma)^{n_{12}} \binom{b}{n_{21}} \gamma^{n_{21}} (1 - \gamma)^{n_{22}} \\ &= \binom{a}{n_{11}} \binom{n - a}{n_{+1} - n_{11}} \gamma^{n_{+1}} (1 - \gamma)^{n_{+2}}. \end{aligned}$$

On the other hand, the probability of the observed column marginals is, again under H_0 ,

$$\binom{n}{n_{+1}} \gamma^{n_{+1}} (1 - \gamma)^{n_{+2}}.$$

Thus, the conditional probability, given the column marginal, is free of γ , and it equals

$$\frac{\binom{a}{n_{11}} \binom{n - a}{n_{+1} - n_{11}}}{\binom{n}{n_{+1}}},$$

which agrees with (15.26).

■ EXAMPLE 15.14

Returning to Example 15.13, the p -value of the observed contingency table would be the same as calculated there (i.e., about 0.12) if we selected six students with high and six with low math scores, and then classified them according to their music scores.

■ EXAMPLE 15.15

Twenty occasional headache sufferers participated in testing a newly developed headache remedy. They were given the drug and later asked whether or not it was significantly better than the drug they usually took. In fact, however, every fourth subject tested received not a drug but a placebo. The data are as follows:

	Significant improvement	No significant improvement	
Drug	11	4	15
Placebo	3	2	5
	14	6	20

Do these data indicate that the new drug is better than the drugs usually taken?

SOLUTION. We are testing here the hypothesis of homogeneity against the alternative of a positive association between taking the new drug and beneficial effects for patients. To determine the p -value, observe that the count n_{11} , given the marginals, may be only

11, 12, 13, or 14 if the table is to be at least as much in favor of the alternative as the observed one. Tables

$$\begin{bmatrix} 11 & 4 \\ 3 & 2 \end{bmatrix}, \begin{bmatrix} 12 & 3 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 13 & 2 \\ 1 & 4 \end{bmatrix}, \begin{bmatrix} 14 & 1 \\ 0 & 5 \end{bmatrix},$$

have probabilities

$$\frac{\binom{15}{11}\binom{5}{3}}{\binom{20}{14}} = 0.352, \quad \frac{\binom{15}{12}\binom{5}{2}}{\binom{20}{14}} = 0.117, \quad \frac{\binom{15}{13}\binom{5}{1}}{\binom{20}{14}} = 0.014, \quad \frac{\binom{15}{14}\binom{5}{0}}{\binom{20}{14}} = 0.0004,$$

respectively. Hence, the p -value is about 0.48, so there is no evidence to back the claim that the new drug is superior to the drugs used so far.

The testing procedure described above is known as the Fisher's exact test (see Section 12.7). In testing the null hypothesis $H_0 : \theta = 1$ against a two-sided alternative $H_1 : \theta \neq 1$, one encounters the usual problem of defining p -values in the case of two-sided alternatives. There seems to be no agreement among statisticians as to what is the proper procedure in such situations. In the case of continuous and symmetric distributions of a test statistic (e.g., Student's t), the p -value for the observed result is usually taken as a doubled p -value for the one-sided alternative. In the case of an asymmetric distribution, there is little justification of doubling the one-sided p -value. In the case of discrete distributions, an additional source of difficulty is that by such a procedure one can obtain a value exceeding 1. Some authors suggest taking as the p -value the sum of all probabilities of tables that are at most as likely as the observed one (see Freeman and Halton, 1951).

In all situations already considered in this section, observations were classified according to two arbitrary variables. There are, however, cases when classification variables are dependent, very often it is the same classification variable applied to data collected on two occasions. For example, one might want to study the effect of a certain medical treatment by comparing data from some tests collected from the same patients before and after the treatment was administered. If the classification variables have dichotomous response (such as 0 and 1, "Yes" and "No," etc.), one can use the so-called *McNemar's test* to compare marginal distributions. The McNemar's test is considered a counterpart of a matched-pair t test discussed in Chapter 12. The data to be analyzed are summarized as in the table below:

Variable 1	Variable 2		
	Yes	No	
Yes	N_{11}	N_{12}	N_{1+}
No	N_{21}	N_{22}	N_{2+}
	N_{+1}	N_{+2}	n

Marginal distributions are (p_{1+}, p_{2+}) and (p_{+1}, p_{+2}) for variable 1 and variable 2, respectively. To test if marginal distributions of classification variables are the same (marginal homogeneity), we will be testing

$$H_0 : p_{1+} = p_{+1} \quad \text{against} \quad H_1 : p_{1+} \neq p_{+1}. \quad (15.27)$$

Hypotheses in (15.27) reduce to hypotheses

$$H_0 : p_{12} = p_{21} \quad \text{and} \quad H_1 : p_{12} \neq p_{21}. \quad (15.28)$$

Of course we could perform the McNemar's test against one-sided alternatives, either

$$H_0 : p_{1+} = p_{+1} \quad \text{against} \quad H_1 : p_{1+} > p_{+1}, \quad (15.29)$$

or

$$H_0 : p_{1+} = p_{+1} \quad \text{against} \quad H_1 : p_{1+} < p_{+1}. \quad (15.30)$$

The measure of the difference between marginal probabilities $p_{1+} - p_{+1}$ reduces to $p_{12} - p_{21}$ and its MLE is the difference of relative frequencies $N_{12}/n - N_{21}/n$. Under the H_0 the distribution of N_{12} is $\text{BIN}(N^*, 1/2)$, where $N^* = N_{12} + N_{21}$. Since it is a symmetric distribution, its normal approximation $N(N^*/2, N^*/4)$ can be applied for $N > 10$. Consequently, the test statistic we then take

$$Z = \frac{N_{12} - N^*/2}{\sqrt{N^*/4}} = \frac{N_{12} - N_{21}}{\sqrt{N_{12} + N_{21}}} \quad (15.31)$$

has approximately $N(0, 1)$ distribution.

■ EXAMPLE 15.16

A student who was not doing well in his statistics class was offered an online tutorial to enhance his algebra skills. To assess the effectiveness of an online tutorial, the student was required to take a pre-test at the beginning and a post-test at the end of the last tutorial session. Both, pre- and post-tests consisted of the same 30 multiple-choice questions so each answer was either correct or incorrect. Student's results are summarized in the table below.

Pre-test	Post-test		
	Correct	Incorrect	
Correct	6	2	8
Incorrect	18	4	22
	24	6	30

Did the student improve his algebra skills by taking the online tutorial?

SOLUTION. Here we will be testing null hypothesis $H_0 : p_{12} = p_{21}$ of no change against one-sided alternative $H_1 : p_{12} < p_{21}$. The value of the test statistic (15.31) is $(2 - 18)/\sqrt{2 + 18} = -3.578$, what corresponds to p -value equal 0.0002. The data provide strong evidence that student's algebra skills improved by taking online tutorial.

At the end of this section, we would like to mention that similarly to all tests that were discussed, we could obtain confidence intervals for various statistics. Unlike the McNemar's test statistic that uses only the information about the counts in different categories for two variables (N_{12} and N_{21}), while the counts N_{11} and N_{22} are irrelevant, the formula for the confidence interval depends on all four counts. Interested readers are advised to check the texts on the analysis of categorical data, for example Agresti (2002).

PROBLEMS

- 15.5.1** Use the data from Problem 15.3.5 to test if high school GPA's are positively related to college GPA's.
- 15.5.2** Two out of ten randomly selected men, and four out of ten randomly selected women were found to be allergic to some drug. **(i)** Does these data indicate that there is a difference between men and women in their propensity to develop an allergic reaction to the drug in question? **(ii)** Does these data indicate that men are less likely than women to develop an allergic reaction to the drug in question? Find corresponding p -values for (i) and (ii) and explain why are they different.
- 15.5.3** A study of change in employment status was conducted among residents of a certain county. A random sample of 100 adult residents was surveyed on their employment status, and their responses (Yes/No) were recorded. The same people were contacted after 12 months and asked the same question. Their answers are summarized in the table below. Does the employment status in the county changed significantly within these 12 months?

Survey 1: Employed	Survey 2: Employed		
	Yes	No	
Yes	80	10	90
No	2	8	10
	82	18	100

- 15.5.4** Fifty randomly selected college students that take courses with on-line quizzes were classified according to their answer on two questions: "Do you cheat taking on-line quizzes?" and "Do other students that you know well cheat taking on-line quizzes?" Explain what can be said about student's cheating. Formulate appropriate hypotheses and perform the test using the data in the table below.

Student	Other students cheat	
	Yes	No
Yes	5	2
No	13	30

15.6 $R \times C$ CONTINGENCY TABLES

At the end, we will extend the results for 2×2 to the general case of $r \times c$ tables, in which the categories corresponding to rows and those corresponding to columns are ordered. The assumption of Theorem 15.5.1 carries over to the present case. By conditioning on both marginals, we obtain (under the null hypothesis of independence or homogeneity) a probability distribution defined on the class of all $r \times c$ tables with the given marginals, which does not involve any unknown parameters.

For the case where both marginals are random, the situation is as follows: Suppose that we take n independent observations, each with the same distribution, and classify them according to two systems with r and c categories, respectively. The cell counts are N_{ij} , and the marginal counts are N_{i+} and N_{+j} , where $i = 1, \dots, r$ and $j = 1, \dots, c$. We have then the following theorem:

Theorem 15.6.1 *Under the assumption of independence, for any contingency table $[n_{ij}]$, $i = 1, \dots, r, j = 1, \dots, s$, with $\sum_{i,j} n_{ij} = n$ and with marginal counts n_{i+} and n_{+j} , we have*

$$P\{[N_{ij}] = [n_{ij}] | N_{i+} = n_{i+}, N_{+j} = n_{+j}\} = \frac{\prod_{i=1}^r (n_{i+})! \prod_{j=1}^c (n_{+j})!}{n! \prod_{i=1}^r \prod_{j=1}^c (n_{ij})!}. \quad (15.32)$$

In order to implement equation (15.22) for the p -value of an observed contingency table, say A_0 , it remains to order all $r \times c$ tables with marginals the same as those of A_0 , according to their “strength of support” for the alternative hypothesis of (say) a positive association.

In presenting the solution, we will use an approach different than that which we used for 2×2 tables. Instead of expressing the null hypothesis in terms of a single parameter and then finding its empirical counterpart (as we did in the case of 2×2 tables), we will introduce several indices (statistics, i.e., functions of the observed counts N_{ij}). Each of these indices will provide, on an intuitive ground, an ordering of contingency tables according to the strength of support for the alternative.

Symbolically, let A_0 be the observed contingency table, and let A be any contingency table with the same marginals as A_0 . Furthermore, let $t = t(A)$ be a real-valued function defined on the considered class of contingency tables. Then, the p -value of the observed table is defined as

$$\sum_{A: t(A) \geq t(A_0)} P_0(A). \quad (15.33)$$

The only problem remaining is the choice of statistic t so that its values could reflect the order of “strength of support” of the alternative hypothesis. The choices here are based on the following idea: Consider a pair of observations. Since each observation is classified as belonging to one of the rows and one of the columns, such a pair determines two pairs of coordinates, say (r', c') and (r'', c'') .

Definition 15.6.1 The pair of observations is called *concordant* if

$$(r'' - r')(c'' - c') > 0;$$

it is called *discordant* if

$$(r'' - r')(c'' - c') < 0,$$

and it is called *tied* if $(r'' - r')(c'' - c') = 0$. □

It should be obvious that every concordant pair provides support in favor of the alternative of positive association: the differences $r'' - r'$ and $c'' - c'$ are both nonzero and of the same sign. This means that a higher evaluation on one variable is accompanied by a higher evaluation on the other variable. By the same logic, every discordant pair provides a support for the alternative of a negative association: a higher classification on one variable is accompanied by a lower evaluation on the other variable.

Let C = number of concordant pairs, and let D = number of discordant pairs. If we can assume that any concordant (or discordant) pair equally supports the alternative, then the overall support for the alternative of (say) a positive association is a function of the difference $C - D$. Standardizing this difference, we let

$$\hat{\gamma} = \frac{C - D}{C + D}. \quad (15.34)$$

The choice of the symbol reflects the fact that $\hat{\gamma}$ (the estimator of some parameter γ) depends on the observed contingency table.

We can now define the p -value of the observed contingency table (in testing the alternative of positive association) as the sum of probabilities of all the tables with the same marginals as the one observed, and with the value of index $\hat{\gamma}$ at least as high as for the original table.

■ EXAMPLE 15.17

For 2×2 tables, the numbers of concordant and discordant pairs of observations are $C = n_{11}n_{22}$ and $D = n_{12}n_{21}$, respectively; hence

$$\hat{\gamma} = \frac{n_{11}n_{22} - n_{12}n_{21}}{n_{11}n_{22} + n_{12}n_{21}} = \frac{\theta - 1}{\theta + 1}, \quad \text{where} \quad \theta = \frac{n_{11}n_{22}}{n_{12}n_{21}}.$$

Notice that the ordering of the values of $\hat{\gamma}$ coincides with the ordering according to the values of θ .

■ EXAMPLE 15.18

For the case of marijuana use in colleges (recall Example 15.12), the categories represented by the rows and those represented by the columns were ordered by a natural way: by the number of parents who use drugs or alcohol and by the frequency of marijuana use by the student.

A concordant pair is formed by observations such that one of them is below and to the right of the other:

$$\begin{aligned} C &= 141 \times (44 + 51 + 11 + 19) + 54 \times (51 + 19) \\ &\quad + 68 \times (11 + 19) + 44 \times 19 = 24,281, \end{aligned}$$

and similarly

$$\begin{aligned} D &= 40 \times (68 + 44 + 17 + 11) + 54 \times (68 + 17) \\ &\quad + 51 \times (17 + 11) + 44 \times 17 = 12,366. \end{aligned}$$

Thus $\hat{\gamma} = 0.325$.

To evaluate the p -value corresponding to an observed contingency table, one should use a statistical software such as R or SAS.

The index $\hat{\gamma}$ utilizes only the information contained in the numbers C and D of concordant and discordant pairs. The numbers of tied pairs are not used, since a pair of observations tied on one or both variables provides evidence neither in favor of the null hypothesis nor in favor of the alternative. However, one could argue that a large number of tied pairs (as compared with $C + D$) is an indication that the difference $C - D$ might be insignificant. This argument has led to the introduction of two indices that take tied pairs into account, called Kendall's tau-b and Somers' d.

The total number of all possible pairs of observations (disregarding the order) is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

Therefore, the total number of tied observations is $n(n-1)/2 - C - D$. Let T_X , T_Y , and T_{XY} denote numbers of pairs of observations tied on the first coordinate (row), on the second coordinate (column), and on both coordinates (falling to the same cell), respectively.

Observe that the categories of ties are not disjoint: any pair tied on both coordinates is counted twice, in T_X and in T_Y .

Definition 15.6.2 The statistic

$$\tau_b = \frac{C - D}{\sqrt{[n(n-1)/2 - T_X][n(n-1)/2 - T_Y]}} \quad (15.35)$$

is called *Kendall's tau-b*. □

Definition 15.6.3 The statistic

$$d = \frac{C - D}{n(n-1)/2 - T_X} \quad (15.36)$$

is called *Somers' d*. □

In the formulas above, we have

$$T_X = \sum_{i=1}^r \frac{n_{i+}(n_{i+} - 1)}{2} \quad \text{and} \quad T_Y = \sum_{j=1}^c \frac{n_{+j}(n_{+j} - 1)}{2}. \quad (15.37)$$

■ EXAMPLE 15.19

Returning to the marijuana data (Examples 15.12 and 15.18), we have $T_X = 41,779$ and $T_Y = 37,306$. The total number of all possible pairs is $445 \times 444/2 = 98,790$. We have here $\tau_b = 0.201$ and $d = 0.209$ (while $\hat{\gamma} = 0.325$).

The calculation of the p -value based on τ_b or d for the alternative of (say) a positive association again uses the sum of all probabilities of tables with the same marginals as those observed, and the index τ_b , or d , at least as high as that for the observed table.

It ought to be mentioned that the very fact of an introduction of an index is—to a certain extent—an attempt to reduce the problem to one dimension. That enables one to compare objects (in this case, contingency tables) and select the best. This may be, however, a deceptive comfort, since not all things are comparable in such a simple way. In the present case, just thinking in terms of an index (e.g., $\hat{\gamma}$) leads to the danger of attaching significance to certain values of that index. For instance, one may tend to take $\hat{\gamma} = 1$ first as an indication of a very strong positive association (which it is), and then tend to attach to it a fixed significance level. To see this, observe that tables

$$\begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 3 & 3 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

both have $D = 0$; hence $\hat{\gamma} = 1$. The p -value for the first table equals

$$\frac{(4!4!)(2!4!2!)}{8!2!2!2!} = 0.086,$$

whereas the p -value for the second table is about half of that for the first table:

$$\frac{(6!6!)(3!5!2!)}{10!3!2!3!2!} = 0.008.$$

This is simply the effect of regarding as equivalent two situations characterized by the same value of $\hat{\gamma}$. This example shows that the values of $\hat{\gamma}$ are not comparable for experiments with different marginal counts.

Computations of the p -value become cumbersome especially when the number of row and/or column categories increases. When cell counts are sufficiently large, normal approximation of sampling distribution may be used (e.g., see Agresti, 2002).

PROBLEMS

15.6.1 A study reported in *Science* magazine investigated the relationship between sex, handedness (right- or left-handed), and relative foot size (left foot bigger than right foot, left foot within one-half shoe size of right foot, or right foot bigger). A random sample of 150 adults gave the following data:

Relative foot size	Right-handed		Left-handed	
	Male	Female	Male	Female
$L > R$	2	55	6	0
$L \approx R$	10	18	6	2
$L < R$	28	14	0	9

Test the association between gender and handedness considering three groups of people: these whose left foot is bigger than their right foot, these whose both feet are almost the same, and these whose left foot is bigger. Compare the results obtained.

15.6.2 Measure the association of GPA and attitude toward statistics courses based on the data from Problem 15.3.5 using $\hat{\gamma}$, τ_b , and d coefficients.